

# RAK12025 WisBlock Gyroscope Sensor Module Datasheet

## Overview

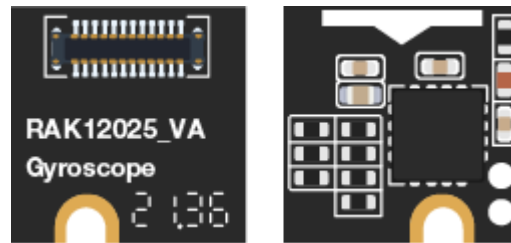


Figure 1: RAK12025 WisBlock Gyroscope Sensor Module

## Description

RAK12025 is a gyroscope module, part of the RAKWireless WisBlock Sensor series. The module is based on I3G4250D from STMicroelectronics. The I3G4250D is a low-power 3-axis angular rate sensor able to provide unprecedented stability at a zero-rate level and sensitivity over temperature and time. It includes a sensing element and a digital interface capable of providing the measured angular rate. With I3G4250D, RAK12025 can measure rotation speed and report data through a standard I2C digital interface.

## Features

- Gyroscope module
- Selectable full scale: (245/500/2000 dps)
- I2C interface
- 16-bit rate value data output
- 8-bit temperature data output
- Module size: 10 × 10 mm
- Power Supply Voltage: 3.3 V
- Current Consumption: 5 uA - 6.1 mA
- Chipset: STMicroelectronics I3G4250D

# Specifications

## Overview

## Mounting

Figure 1 shows the mounting mechanism of the RAK12025 module on a [WisBlock Base](#) board. The RAK12025 module can be mounted on the slots: C, D, E, & F.

### NOTE

RAK12025 has two digital output lines, so you need two GPIOs from WisBlock Core. It means RAK12025 should be used on a sensor slot with two available GPIOs. However, WB\_IO2 is used to control 3V3\_S. Hence, RAK12025 is used only on slots without WB\_IO2 like sensor slots C and D on WisBlock Base board.

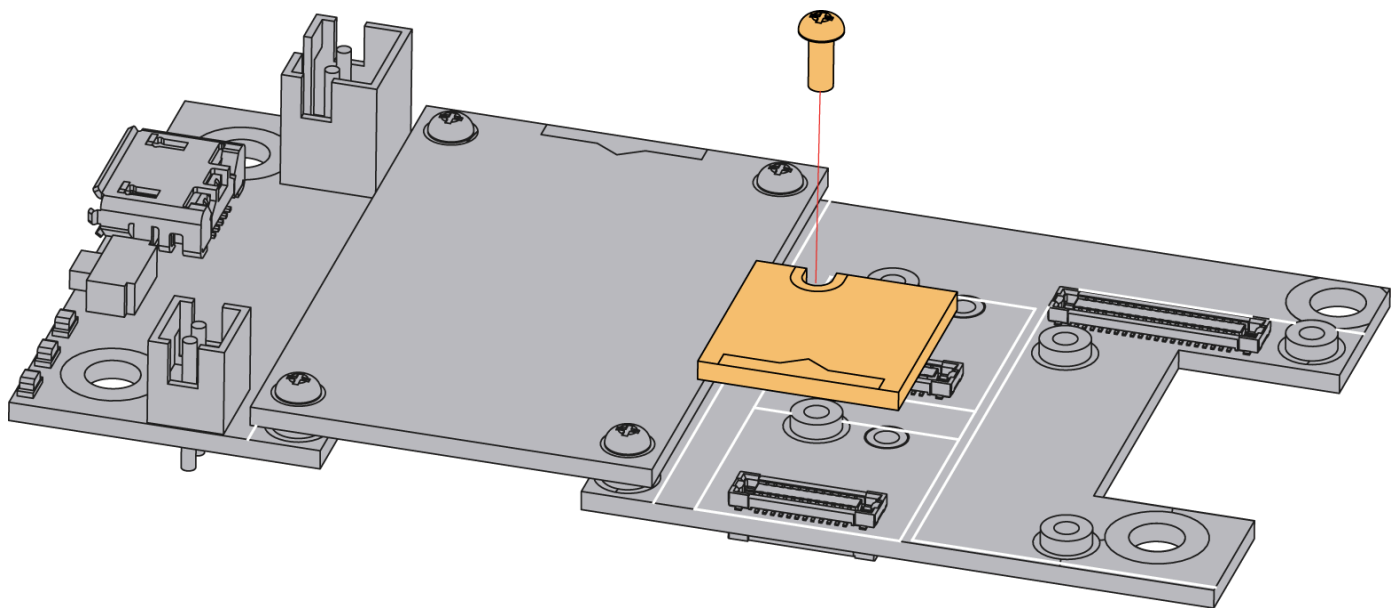


Figure 2: Mounting of RAK12025

## Hardware

The hardware specification is categorized into six (6) parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12025 WisBlock Gyroscope Sensor Module.

## Chipset

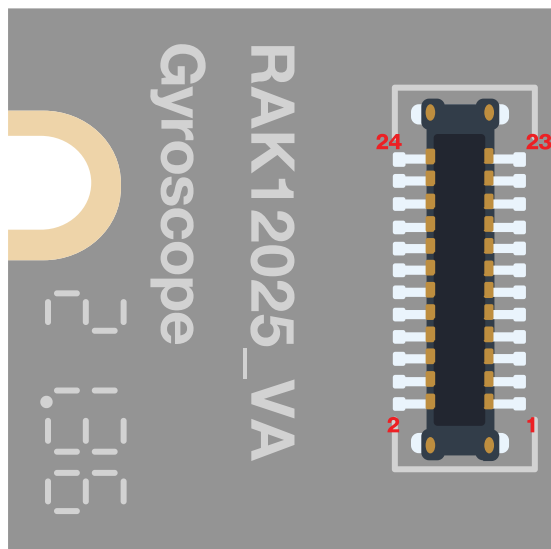
| Vendor             | Part number |
|--------------------|-------------|
| STMicroelectronics | I3G4250D    |

## Pin Definition

The RAK12025 WisBlock Gyroscope Sensor Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK12025 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 3**.

### NOTE

I2C related pin, INT2, INT1, 3V3\_S, and GND are connected to Wisblock connector.



|         |    |    |         |
|---------|----|----|---------|
| GND     | 2  | 1  | NC      |
| NC      | 4  | 3  | NC      |
| NC      | 6  | 5  | NC      |
| I2C_SDA | 8  | 7  | I2C_SCL |
| INT1    | 10 | 9  | NC      |
| INT2    | 12 | 11 | 3V3_S   |
| 3V3_S   | 14 | 13 | INT2    |
| NC      | 16 | 15 | INT1    |
| I2C_SCL | 18 | 17 | I2C_SDA |
| NC      | 20 | 19 | NC      |
| NC      | 22 | 21 | NC      |
| NC      | 24 | 23 | GND     |

Figure 3: RAK12025 pinouts

If a 24-pin WisBlock Sensor connector is used, the IO used for the output pulse depends on what slot the module is plugged in. The following table shows the default IO used for different slots:

The table below shows the default IOs used for different slots using INT1:

| SLOT C | SLOT D | SLOT E | SLOT F |
|--------|--------|--------|--------|
| WB_IO4 | WB_IO6 | WB_IO3 | WB_IO5 |

When using INT2, the default IOs used for different slots is shown in the table below:

| SLOT C | SLOT D | SLOT E | SLOT F |
|--------|--------|--------|--------|
| WB_IO3 | WB_IO5 | WB_IO4 | WB_IO6 |

## Sensors

### Gyroscope Sensor

| Symbol | Parameter         | Test Condition   | Min. | Typ.  | Max. | Unit       |
|--------|-------------------|------------------|------|-------|------|------------|
| FS     | Measurement Range | User-selectable  |      | ±245  |      | dps        |
|        |                   | User-selectable  |      | ±500  |      | dps        |
|        |                   | User-selectable  |      | ±2000 |      | dps        |
| So     | Sensitivity       | FS bit = 245 dps | 7.4  | 8.75  | 10.1 | mdps/digit |
|        |                   | FS bit = 500 dps | 14.8 | 17.50 | 19.8 | mdps/digit |

| Symbol | Parameter                                       | Test Condition               | Min.   | Typ.  | Max.   | Unit       |
|--------|-------------------------------------------------|------------------------------|--------|-------|--------|------------|
|        |                                                 | FS bit =<br>2000 dps         | 59.2   | 70    | 79.3   | mdps/digit |
| SoDr   | Sensitivity<br>change vs.<br>Temperature        | From<br>-40 °C to<br>+85 °C  |        | ±2    |        | %          |
| DVoff  | Digital zero-<br>rate level                     | FS =<br>245 dps              | -25    | ±10   | +25    | dps        |
|        |                                                 | FS =<br>500 dps              | -37.5  | ±15   | +37.5  | dps        |
|        |                                                 | FS =<br>2000 dps             | -187.5 | ±75   | +187.5 | dps        |
| OffDr  | Zero-rate<br>level change<br>vs.<br>Temperature | FS =<br>245 dps              |        | ±0.03 |        | dps/°C     |
| NL     | Non-linearity                                   | Best fit<br>straight<br>line | -5     | 0.2   | +5     | dps/°C     |
| DST    | Self-test<br>output<br>change                   | FS =<br>245 dps              |        | 130   |        | dps        |
|        |                                                 | FS =<br>500 dps              |        | 200   |        | dps        |

| Symbol | Parameter                   | Test Condition   | Min. | Typ.            | Max. | Unit          |
|--------|-----------------------------|------------------|------|-----------------|------|---------------|
|        |                             | FS =<br>2000 dps |      | 530             |      | dps           |
| Rn     | Rate noise density          | BW =<br>50 Hz    |      | 0.03            |      | dps/sqrt (Hz) |
| ODR    | Digital output data rate    |                  |      | 105/208/420/840 |      | Hz            |
| Top    | Operating temperature range |                  | -40  |                 | +85  | °C            |

## Electrical Characteristics

- @ Vdd = 3.0 V, T = +25 °C, unless otherwise noted.

| Symbol | Description                       | Condition                            | Min. | Typ. | Max. | Unit |
|--------|-----------------------------------|--------------------------------------|------|------|------|------|
| Vdd    | Supply voltage                    | Input voltage must within this range | 2.4  | 3.3  | 3.6  | V    |
| Idd    | Supply current                    | Normal                               | -    | 6.1  | -    | mA   |
| IddSL  | Supply current in sleep mode      | Sleep mode                           | -    | 1.5  | -    | uA   |
| IddPdn | Supply current in power-down mode | Power-down mode                      |      | 5    |      | uA   |

| Symbol | Description                 | Condition | Min. | Typ. | Max. | Unit |
|--------|-----------------------------|-----------|------|------|------|------|
| Top    | Operating temperature range |           | -40  |      | +85  | °C   |

## Mechanical Characteristics

### Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12025 module.

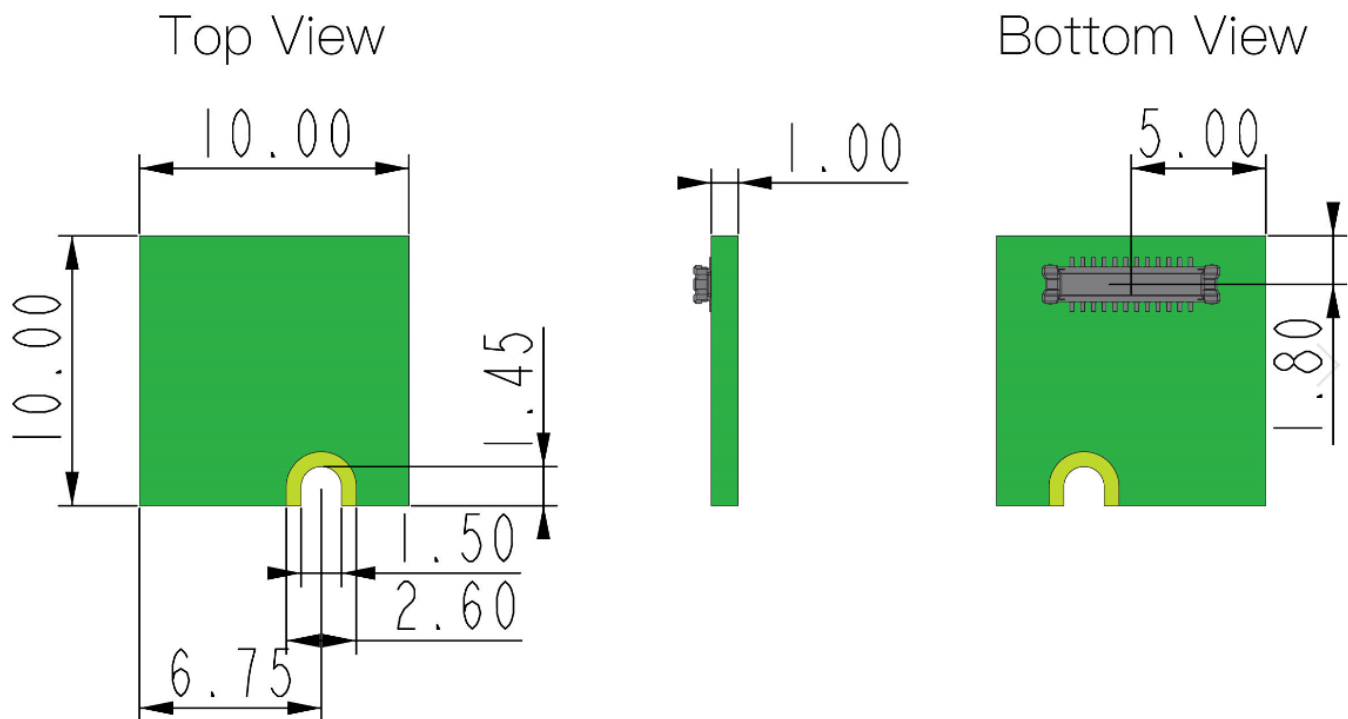
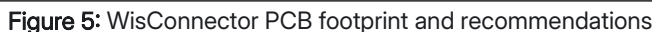


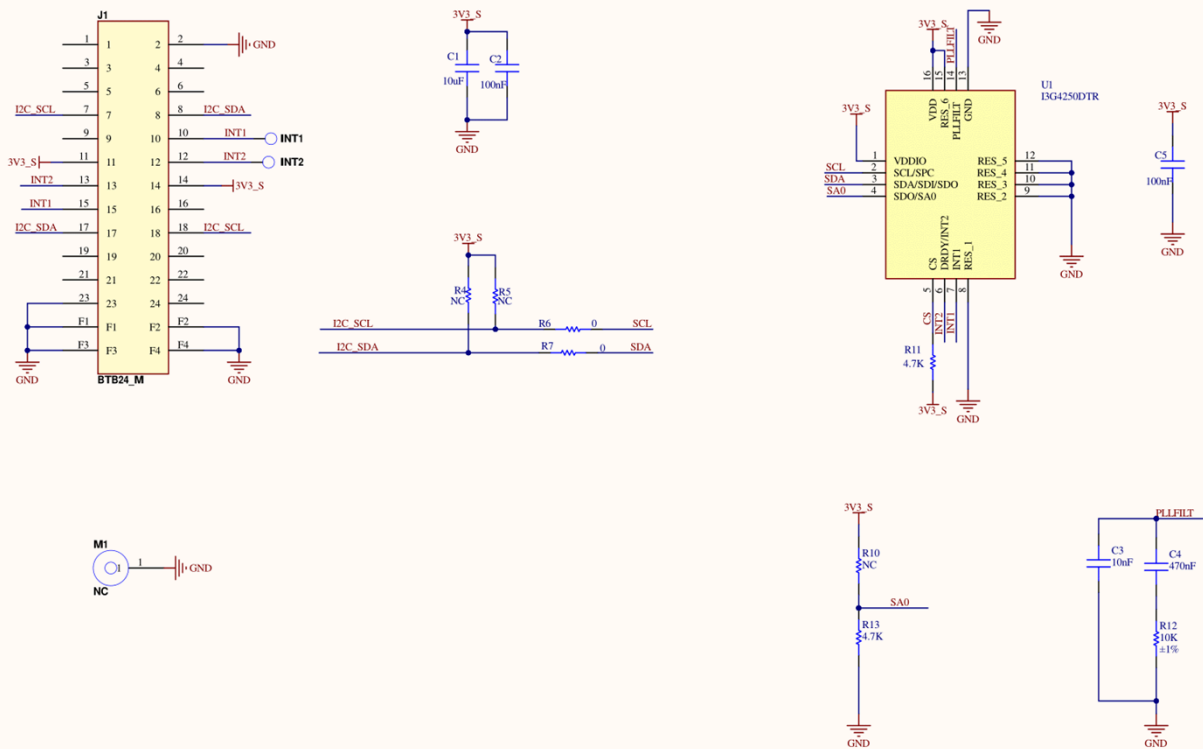
Figure 4: RAK12025 mechanical dimensions

## WisConnector PCB Layout



### Schematic Diagram





The slave address (7 bit) :1101000

Figure 6: RAK12025 schematic diagram